

Subsystem Specification Helm

Navigate, enforce the fence and run every native failsafe on an independent RTOS processor, whatever else has failed.

Document	BOATY-SSS-HLM
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Status	Baselined at the System Design Review (tag sdr-baseline-1); changes only by change request
Parents	SRS Issue H, ADD Issue H, ICD Issue H, FMEA Issue G, KCL Issue D
Content	35 allocated SRS requirements → 44 subsystem requirements (39 M, 3 S, 2 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)
B	28 September 2026	CR-02: microSD logging (HLM-D02). FMEA actions A-01, A-02, A-06, A-18: HLM-D24 formalised; HLM-D39 to D41 added; parameter baseline extended.	Claude, owner decisions (CR-02, FMEA actions)
C	28 September 2026	CR-03: board class and real price (Matek F405-TE class, £62) in budget and HLM-D02.	Claude, owner decision
D	29 September 2026	CR-04: SpeedyBee F405 WING APP (HLM-D02). CR-05: GCS failsafe 3 s (HLM-D27). Simulator results folded into HLM-D05/08/20/24/25/26/28/29; HLM-D42 to D44 added; parameter baseline now generated from software/params.	Claude, owner decisions
E	30 September 2026	SDR WP2: HLM-D19 met by Dijkstra path planning with slide avoidance (OA_TYPE, OA_MARGIN_MAX, AVOID_BEHAVE; FENCE_MARGIN 3 → 2 m), verified by SC-27. HLM-D09: HDOP gate moved to Mission Control (not available in Rover).	Claude
F	30 September 2026	SDR WP5 (RID-08): issue, parents and applicable documents taken from the baseline register; issued as the SDR baseline candidate.	Claude

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Navigate, enforce the fence and run every native failsafe on an independent RTOS processor, whatever else has failed.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• Flight controller hardware (SpeedyBee F405 WING APP + PDB) • ArduPilot Rover firmware (third party) and the controlled parameter set • GNSS + compass module and mast cabling • Fence and mission storage • Dataflash / SD logging	• Everything that sends commands (MCN via IF-02, MCP via IF-04) • Motor drives (PRP) • Power sources (PWR)

1.3 Breakdown

Item	Description
HLM-1 Flight controller	SpeedyBee F405 WING APP (STM32F405, ChibiOS), PDB with V/I sense, microSD
HLM-2 Firmware	ArduPilot Rover 4.7.1, pinned
HLM-3 Parameter set	software/params/boaty-mk1.parm (behaviour, also flown in SITL) + boaty-mk1-speedybee.parm (board wiring)
HLM-4 GNSS/compass	u-blox M10 + QMC5883L on the mast

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue H	Parent requirements
BOATY-ADD-001 Issue H	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue H	Interface definitions (all listed in section 3 apply in full)
BOATY-FMEA-001 Issue G	Failure modes and actions carried into this issue
BOATY-KCL-001 Issue D	Component data, parameters and cost

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-02 Command & telemetry	Mission Control	Conforms	\$4
IF-04 Companion link	Mission computer	Conforms	\$4
IF-05 Motor commands	Propulsion	Owner	\$6
IF-07 Helm power & sensing	Power	Conforms	\$6
IF-09 Beacon drive	Recovery & signalling	Owner	\$6
IF-19 Mast socket	Hull & structure	Conforms	\$8
IF-22 GNSS & compass	internal	Owner	\$6

4. Parent requirements

Primary (35): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
MOD-001	M	The system shall implement exactly these boat modes: DISARMED, HOLD, MANUAL, AUTO and RTL, as defined in section 2.3.	HLM-D05
MOD-002	M	On power-up the boat shall enter DISARMED, with both motors inhibited.	HLM-D04
MOD-004	M	The system shall refuse to arm unless every pre-arm check (PRE-001 to PRE-008) passes, and shall state the failing...	HLM-D08
MOD-005	M	When an AUTO mission completes, the boat shall return home inside the fence, avoiding exclusions (RTL behaviour,...	HLM-D06
MOD-008	C	The boat shall disarm automatically after 60 s in HOLD at home with no operator input, and announce it.	HLM-D07
PRE-001	M	GNSS shall have a 3D fix with at least 8 satellites and HDOP no greater than 1.5.	HLM-D09
PRE-002	M	A geofence shall be loaded and enabled, and home shall lie inside the inclusion fence and outside all exclusion zones.	HLM-D10
PRE-003	M	Battery state of charge shall be at least 80%.	HLM-D11
PRE-004	M	The heading sensor shall report healthy and calibrated.	HLM-D12
PRE-007	M	The independent safety layer (SAF-001) shall report healthy.	HLM-D13
PRE-008	M	Logging shall be active on the helm and Mission Control, each with at least 1 GB free.	HLM-D02, HLM-D14
NAV-002	M	Cruise speed in calm water shall be 1.0 m/s $\pm$ 0.2 m/s.	HLM-D15
NAV-003	M	Speed through the water shall be limited to 1.5 m/s in every mode.	HLM-D16
NAV-004	M	In AUTO the boat shall pass within the acceptance radius of each waypoint. The radius is configurable from 1 to 10 m,...	HLM-D17
NAV-005	S	Cross-track error between waypoints shall be no more than 3 m RMS in wind up to 3 m/s.	HLM-D18, HLM-D42
NAV-006	S	In HOLD with station-keeping, the boat shall stay within 5 m of the hold point in wind up to 3 m/s.	HLM-D18
NAV-007	M	In RTL the boat shall navigate to home without entering any exclusion zone or leaving the inclusion fence.	HLM-D19
NAV-008	S	Above 0.3 m/s the heading estimate shall use GNSS course as well as the magnetometer, so a disturbed compass cannot...	HLM-D20
FEN-001	M	The system shall support one inclusion polygon of 3 to 70 vertices and up to 10 exclusion zones, each a polygon or...	HLM-D10, HLM-D21
FEN-004	M	The fence shall be enforced in every armed mode, including MANUAL.	HLM-D22
FEN-005	M	On a fence breach the boat shall enter RTL within 1 s of the breach.	HLM-D23
FEN-007	M	Fence and exclusion zones shall not be changed while the boat is armed.	HLM-D25
FS-001	M	Low battery: when estimated state of charge falls to 35%, the boat shall enter RTL. At 15% it shall continue RTL at...	HLM-D26, HLM-D40
FS-002	M	Link loss in MANUAL: with no command link for 3 s the boat shall enter HOLD, and after 10 s it shall enter RTL. (3 s...	HLM-D27
FS-003	M	Link loss in AUTO: the boat shall continue the mission. If the link is still lost after 60 s it shall enter RTL.	HLM-D27
FS-005	M	Stuck detection: if commanded thrust is at least 50% and ground speed stays below 0.1 m/s for 5 s, the boat shall...	HLM-D29
FS-007	M	Mission computer failure: if the helm receives no mission computer heartbeat for 3 s, it shall complete the loaded...	HLM-D30

SRS	Pri	Requirement (abridged)	Met by
FS-011	M	When several failsafes are active, the most conservative response shall win: motors stopped over RTL over continuing...	HLM-D31
FS-012	M	Every failsafe event shall be logged, and announced at Mission Control within 2 s if the link is up.	HLM-D32
SAF-001	M	Fence and failsafe functions shall run on a processor separate from the mission computer and Mission Control. That...	HLM-D01, HLM-D02, HLM-D03
SAF-002	M	Those functions shall keep working with the mission computer, Mission Control and the radio link all failed.	HLM-D30, HLM-D33
SAF-003	M	The mission computer and Mission Control shall command the helm only through the helm interface (SWE-002). They shall...	HLM-D34, HLM-D44
COM-007	C	An independent manual-override radio link (e.g. ExpressLRS) may be added.	HLM-D37
MEC-013	M	The GNSS and compass shall be mounted at least 150 mm from motor and power wiring.	HLM-D35, HLM-D41
LOG-001	M	The helm shall log position, attitude, mode, commands, battery and failsafe events at 5 Hz or faster.	HLM-D36

Supporting (24): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

MOD-003, MOD-006, PRE-005, NAV-001, FEN-006, FS-004, FS-008, FS-009, FS-010, SAF-007, MIS-001, MIS-002, MIS-003, MIS-004, MIS-005, MIS-006, DET-004, MC-014, REC-004, PWR-005, PWR-006, PWR-010, SWE-008, ENV-002

5. Subsystem requirements

Requirements are numbered HLM-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Hardware and firmware

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D01	The helm shall run a pinned ArduPilot Rover stable release built for the chosen board: Rover 4.7.1 (tag Rover-4.7.1, commit db79216), the release the simulator flies. Version and hash are recorded in the repository.	M	I	BENCH	SAF-001, SWE-006
HLM-D02	The flight controller shall provide: ≥ 2 free UARTs (GNSS, companion), ≥ 3 DShot-capable outputs, ≥ 2 analogue battery inputs, I2C, IMU and baro, and a microSD slot (≥ 8 GB card) for logging (CR-02), and run from ≥ 7 V so it survives pack sag near empty. Baseline: SpeedyBee F405 WING APP (7-36 V, 5 usable UARTs, 12 DShot outputs, microSD; CR-04), wireless module not fitted.	M	I	BENCH	SAF-001, PRE-008, V-01, DD-15, DD-19
HLM-D03	No project-written code (including Lua) shall run on the flight controller. It runs ChibiOS only.	M	I	BENCH	SAF-001, DD-13
HLM-D04	On power-up the helm shall be disarmed with motor outputs at neutral. Auto-arming shall be disabled.	M	T	BENCH	MOD-002

5.2 Modes

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D05	Only STEERING, HOLD, LOITER, AUTO, RTL and GUIDED (weed-shedding only) shall be used. The helm shall boot and arm into HOLD (INITIAL_MODE 4), with no RC mode switch (MODE_CH 0) and any RC receiver ignored (RC_OPTIONS 1). Without these SITL armed into MANUAL.	M	T	SIM	MOD-001, DD-12
HLM-D06	At mission end the helm shall RTL and then hold at home. The mission-done behaviour is set to hold as a backstop.	M	S	SIM	MOD-005, V-04
HLM-D07	Auto-disarm after 60 s at home is performed by Mission Control (MCN-D12). The helm shall accept it.	C	S	SIM	MOD-008

5.3 Pre-arm checks

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D08	All ArduPilot arming checks shall be enabled except the RC-channels check (no receiver is fitted: ARMING_SKIPCHK 64), and failures reported as STATUSTEXT (MCN translates them).	M	S	SIM	MOD-004
HLM-D09	Arming shall need a 3D fix and a healthy EKF. Rover has no HDOP arming gate (GPS_HDOP_GOOD is Copter-only; SDR RID-11), so the HDOP ≤ 1.5 and ≥ 8 satellite checks are done by Mission Control (MCN-D15).	M	T	BENCH	PRE-001
HLM-D10	Arming shall need the fence enabled and loaded, with home inside it.	M	S	SIM	PRE-002, FEN-001
HLM-D11	Arming shall need $\geq 80\%$ battery capacity remaining.	M	S	SIM	PRE-003
HLM-D12	Arming shall need calibrated, consistent compass and EKF.	M	T	BENCH	PRE-004
HLM-D13	Arming shall be refused with motor-rail voltage < 9.0 V (key out).	M	T	BENCH	PRE-007, MOD-003, V-13
HLM-D14	Arming shall need logging available.	M	T	BENCH	PRE-008

5.4 Navigation

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D15	Cruise speed shall be set to 1.0 m/s and tuned to ± 0.2 m/s.	M	T	POOL	NAV-002
HLM-D16	Waypoint and RTL speeds shall be ≤ 1.5 m/s. Maximum throttle shall be limited so full throttle in STEERING gives ≤ 1.5 m/s in calm water.	M	T	POOL	NAV-003
HLM-D17	Waypoint acceptance radius shall default to 3 m (configurable 1-10 m).	M	S	SIM	NAV-004
HLM-D18	Steering and speed controllers shall be tuned to ≤ 3 m RMS cross-track error and ≤ 5 m loiter error in 3 m/s wind.	S	T	LAKE	NAV-005, NAV-006
HLM-D19	RTL shall plan around exclusion zones inside the inclusion fence: ArduPilot's Dijkstra planner (OA_TYPE 2, OA_MARGIN_MAX 6 m) with fence avoidance set to slide along a zone's margin (AVOID_BEHAVE 0, FENCE_MARGIN 2 m). With Rover's default 'stop' behaviour the planned path stalled the boat (SC-27, SDR RID-01).	M	S, T	SIM	NAV-007, V-01
HLM-D20	NAV-008 shall be met by the helm's EKF together with B7's first-motion heading check (MCP-D22). V-16 showed the EKF alone keeps a wrong heading with its single compass rotated 90°, though navigation still tracked on GNSS course.	S	S	SIM	NAV-008, V-16

5.5 Fence

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D21	The helm shall hold one inclusion polygon (≤ 70 vertices) and ≤ 10 exclusion polygons or circles, plus a 100 m circle backstop about home at Milton.	M	S	SIM	FEN-001, OPS-006
HLM-D22	The fence shall be enforced in every armed mode used.	M	S	SIM	FEN-004, V-02
HLM-D23	A breach shall trigger RTL within 1 s.	M	S	SIM	FEN-005
HLM-D24	Persistent breach (> 30 s or > 10 m outside) shall stop the motors. V-14 found no native mechanism, so B7 (MCP-D30) and Mission Control (MCN-D59) each independently command HOLD, and either suffices. The helm's fence RTL stays the primary response.	M	S	SIM	FEN-006, V-14, A-18, FM-14, DD-21
HLM-D25	Fence changes while armed shall be refused by every sender: C7 (IF-14) and the IF-04 filter. V-17 showed the helm itself accepts them, so these two are the only enforcement.	M	S	SIM	FEN-007, V-17

5.6 Failsafes

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D26	Battery failsafe: RTL at 35% remaining; at 15%, continue RTL and alarm. V-15 found no native speed reduction, so B6 lowers the RTL speed (MCP-D29).	M	S	SIM	FS-001, V-15
HLM-D27	GCS failsafe on system-255 heartbeats: FS_GCS_TIMEOUT 2 s + FS_TIMEOUT 1 s = HOLD 3 s after the link goes (CR-05; both are the firmware minimums), continue in AUTO. The later RTL steps are done by MCP B4 (MCP-D12).	M	T	SIM	FS-002, FS-003, V-03, V-06, DD-22
HLM-D28	EKF/position failsafe $\rightarrow$ HOLD as a backstop. It took 9 s in SITL, so B6 stops the motors within FS-004's 3 s (MCP-D28) and does the recovery RTL (MCP-D15).	M	S	SIM	FS-004, DD-21
HLM-D29	Crash/stuck check $\rightarrow$ HOLD (CRASH_THR_MIN 50%, CRASH_VEL_MIN 0.1 m/s, CRASH_TIMEOUT 5 s), raising an event the MCP acts on. One noisy GNSS speed sample resets its timer (V-05), so B7's second detector backs it up.	M	S	POOL	FS-005, V-05
HLM-D30	No companion-computer failsafe shall be configured: the helm completes missions without the MCP.	M	S	SIM	FS-007, SAF-002
HLM-D31	When failsafes combine, motors-stopped outranks RTL, which outranks continue. Demonstrated in SITL combinations.	M	S	SIM	FS-011

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D32	Every failsafe shall be logged and reported by STATUSTEXT.	M	S	SIM	FS-012

5.7 Independence, identity and logging

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D33	All fence and failsafe functions shall work with the MCP, Mission Control and the link all absent.	M	S	SIM	SAF-002
HLM-D34	Only system 255 is the GCS. Parameter writes are accepted only while disarmed through IF-14. Boat services are filtered (IF-04).	M	T	SIM	SAF-003, IF-02, IF-04
HLM-D35	The GNSS/compass shall be mast-mounted ≥ 150 mm from power wiring, with motor interference $\leq 30\%$ after a motor-interference calibration.	M	T	BENCH	MEC-013
HLM-D36	The helm shall log position, attitude, mode, commands, battery and failsafes at ≥ 5 Hz.	M	T	BENCH	LOG-001
HLM-D37	A spare UART shall be kept for an optional ExpressLRS receiver, with RC failsafe = RTL if fitted.	C	I	BENCH	COM-007
HLM-D38	The beacon output shall be configured per IF-09.	S	T	BENCH	IF-09, REC-004

5.8 FMEA-driven (Issue B)

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D39	GNSS glitch rejection (EKF3, glitch radius ≤ 25 m) and GNSS data-timeout detection shall be enabled. A glitch or frozen position shall not produce an uncommanded exit from the fence (SC-20, SC-21, L2-04).	M	S	SIM	FS-004, FS-013, A-01, FM-02, FM-03
HLM-D40	Voltage thresholds shall back up the mAh battery failsafe: low 10.2 V and critical 9.6 V under load for ≥ 10 s, with the same actions. Whichever triggers first applies.	M	T	BENCH	FS-001, A-06, FM-15
HLM-D41	Compass interference shall be measured on rig L2 at full thrust before the hull layout is frozen. Above 30%, wiring or mast height changes before build.	M	T	BENCH	MEC-013, A-02, FM-04

5.9 Found in simulation (Issue D)

ID	Requirement	Pri	Ver	Stage	Trace
HLM-D42	The throttle-to-thrust curve shall be linearised (MOT_THST_EXPO), because props give thrust $\sim \text{throttle}^2$. At the default a 180° pivot took 20 s in SITL; at 1.0 it takes 6 s. Re-tuned from thrust-stand data before the pool.	M	T	POOL	NAV-001, NAV-005
HLM-D43	Motor power shall be limited (BATT_WATT_MAX 70 W) so pack sag near empty cannot brown out the ESCs or flight controller (KCL section 5), while leaving thrust to make headway in the ENV-002 wind.	M	A	BENCH	PWR-005, DD-19
HLM-D44	The helm shall have exactly one MAVLink command path: SERIAL1 to the mission computer. The flight controller's wireless module is not fitted and SERIAL6 is disabled.	M	I	BENCH	SAF-003, IF-02, IF-04

6. Controlled parameter baseline

The configuration that implements the requirements above. **This table is generated from software/params/boaty-mk1.parm (behaviour, also flown in the simulator) and boaty-mk1-speedybee.parm (board wiring)**, so it is always the file that is flashed. Every behaviour parameter was checked against Rover 4.7.1 in SITL (IF-14-02).

Group	Parameter	Value	Why
Vehicle	FRAME_CLASS	2	Boat (MOD-001, V-01)
Motors: skid steer, outputs 1 and 4	SERVO1_FUNCTION	73	ThrottleLeft
Motors: skid steer, outputs 1 and 4	SERVO2_FUNCTION	0	
Motors: skid steer, outputs 1 and 4	SERVO3_FUNCTION	0	
Motors: skid steer, outputs 1 and 4	SERVO4_FUNCTION	74	ThrottleRight
Motors: skid steer, outputs 1 and 4	MOT_THR_MAX	100	model top speed ~1.5 m/s (NAV-003); re-tune in pool
Motors: skid steer, outputs 1 and 4	MOT_THST_ASYM	2.67	2.0 N ahead / 0.75 N astern per pod (PRP-D02)
Motors: skid steer, outputs 1 and 4	MOT_THST_EXPO	1.0	props: thrust ~ throttle^2. At 0 a 180 deg pivot took 20 s in SITL, at 1.0 it takes 6 s. Re-tune from thrust-stand data (KCL DS-09).
Motors: skid steer, outputs 1 and 4	MOT_REV_DELAY	0.2	s, ESC reversal pause (KCL KC-03; tune on L2)
Motors: skid steer, outputs 1 and 4	MOT_BAT_WATT_TC	2	s, power-limit time constant
Speeds and navigation	CRUISE_SPEED	1.0	m/s (NAV-002)
Speeds and navigation	CRUISE_THROTTLE	70	% for 1.0 m/s in the model; measure in the pool
Speeds and navigation	WP_SPEED	1.0	m/s (NAV-004)
Speeds and navigation	WP_RADIUS	3	m
Speeds and navigation	RTL_SPEED	0	0 = WP_SPEED (NAV-007)
Speeds and navigation	LOIT_RADIUS	2	m (NAV-006)
Speeds and navigation	MIS_DONE_BEHAVE	0	Hold at mission end (MOD-005 backstop)
Speeds and navigation	INITIAL_MODE	4	arm into HOLD, never MANUAL (found in SITL)
Speeds and navigation	MODE_CH	0	no RC mode switch: it overrode INITIAL_MODE
Speeds and navigation	RC_OPTIONS	1	ignore any RC receiver: none is fitted (IF-02)
Identity	MAV_SYSID	1	
Identity	MAV_GCS_SYSID	255	only Mission Control counts as the GCS (V-06)
Fence	FENCE_ENABLE	1	
Fence	FENCE_TYPE	6	circle (2) + polygon (4)
Fence	FENCE_ACTION	1	RTL or Hold

Group	Parameter	Value	Why
Fence	FENCE_RADIUS	100	m backstop circle (OPS-006)
Fence	FENCE_MARGIN	2	m: fence avoidance keeps this clear. Below the validator's 3 m leg clearance, and well below OA_MARGIN_MAX (SC-27: at 3 m it stalled RTL)
Path planning round exclusions	OA_TYPE	2	Dijkstra
Path planning round exclusions	AVOID_BEHAVE	0	slide along a zone's margin, don't stop. The planner's legs can pass close to a zone; with Rover's default (stop) RTL deadlocked (SC-27)
Path planning round exclusions	OA_MARGIN_MAX	6	m round zones for planned paths. Must exceed FENCE_MARGIN plus the ~2 m the boat cuts each corner, or fence avoidance stalls the boat on the planned path (SC-27; ArduPilot #11601)
Battery	BATT_MONITOR	4	analogue voltage + current
Battery	BATT_CAPACITY	2500	mAh; PWR-D20 minimum, replace with measured
Battery	BATT_LOW_MAH	875	35% remaining -> RTL
Battery	BATT_CRT_MAH	375	15% remaining -> RTL
Battery	BATT_FS_LOW_ACT	1	RTL
Battery	BATT_FS_CRT_ACT	1	RTL
Battery	BATT_ARM_MAH	2000	80% needed to arm (PRE-003)
Battery	BATT_LOW_VOLT	10.2	V, voltage backstop (HLM-D40)
Battery	BATT_CRT_VOLT	9.6	V
Battery	BATT_FS_VOLTSRC	1	sag-compensated voltage (KCL KF-01)
Battery	BATT_WATT_MAX	70	W, keeps the rail above 7.5 V near empty (KF-02)
Motor-rail sense: arming key	BATT2_MONITOR	3	analogue voltage only
Motor-rail sense: arming key	BATT2_VOLT_PIN	15	AIRSPD pad on the SpeedyBee; voltage2 in SITL
Motor-rail sense: arming key	BATT2_ARM_VOLT	9.0	V: no arming with the key out
Failsafes	FS_GCS_ENABLE	2	enabled, continue mission in AUTO (FS-002/003)
Failsafes	FS_GCS_TIMEOUT	2	s without a system-255 heartbeat (minimum 2)
Failsafes	FS_TIMEOUT	1	s the failsafe must persist (minimum 1). Total ~3 s: FS-002 relaxed to 3 s (CR-05).
Failsafes	FS_ACTION	2	Hold
Failsafes	FS_THR_ENABLE	0	no RC receiver fitted (manual is via IF-02)
Failsafes	FS_EKF_ACTION	1	Hold (FS-004)
Failsafes	FS_CRASH_CHECK	1	Hold (FS-005)
Failsafes	CRASH_THR_MIN	50	% throttle ...
Failsafes	CRASH_VEL_MIN	0.1	... with speed below this (FS-005) ...
Failsafes	CRASH_TIMEOUT	5	... for this long -> Hold (FS-005: <= 5 s)
Failsafes	EK3_GLITCH_RAD	25	m (FS-004, HLM-D39)
Arming	ARMING_SKIPCHK	64	skip only the RC-channels check: no RC receiver
Board wiring	SERIAL1_PROTOCOL	2	MAVLink 2 to the Pi Zero (IF-04)
Board wiring	SERIAL1_BAUD	115	
Board wiring	SERIAL2_PROTOCOL	-1	unused (SBUS inverter on RX)
Board wiring	SERIAL3_PROTOCOL	5	GPS (IF-22)

Group	Parameter	Value	Why
Board wiring	SERIAL6_PROTOCOL	-1	wireless board not fitted (KCL KF-04)
Board wiring	MOT_PWM_TYPE	6	DShot300 (IF-05)
Board wiring	SERVO_DSHOT_ESC	1	AM32
Board wiring	SERVO_BLH_3DMASK	9	outputs 1 and 4 reversible
Board wiring	BATT_VOLT_PIN	10	
Board wiring	BATT_CURR_PIN	11	
Board wiring	BATT_VOLT_MULT	11.05	calibrate on the bench (KF-10)
Board wiring	BATT_AMP_PERVLT	50	calibrate on the bench (KF-10)
Board wiring	BATT2_VOLT_MULT	11.0	10k / 1k divider (IF-06)
Board wiring	LOG_BACKEND_TYPE	1	microSD
Board wiring	NTF_LED_TYPES	257	board LEDs + NeoPixel
Board wiring	SERVO12_FUNCTION	120	NeoPixel1 on the LED pad (IF-09)
Board wiring	GPS_TYPE	1	auto
Board wiring	COMPASS_EXTERNAL	1	

6.1 Where each failsafe step lives

Requirement	Helm (native)	Boat Python (MCP)	Mission Control
FS-001	RTL at 35%; RTL + alarm at 15%	Slow RTL to 0.6 m/s at 15% (B6)	Announce
FS-002	HOLD at 3 s (CR-05)	RTL at 10 s (B4)	Show 'link lost'
FS-003	Continue in AUTO	RTL at 60 s (B4)	-
FS-004	EKF failsafe HOLD (backstop, ~9 s)	HOLD within 3 s; RTL after 10 s healthy (B6)	Announce
FS-005/006	HOLD on stuck (crash check)	Second detector at 10 s (B7); ≤ 3 astern bursts (B5)	Announce
FEN-005/006	RTL on breach	HOLD at 30 s or 10 m outside (B7)	HOLD if persistent (MCN-D59)

Rule: nothing outside the helm is the last line of defence for an action the helm can take. Where the helm cannot (FS-001 speed, FS-004 timing, FEN-006) the boat services add the step, and each is proven in simulation (software/results).

7. Design constraints and budgets

- ADD DD-19 (CR-04): SpeedyBee F405 WING APP, 7-36 V input
- ADD DD-13: no Lua; Python on the MCP covers Lua-type behaviours
- No project-written code on the FC (SAF-001)

Budget	Allocation
Mass allocation	≤ 45 g (estimate 39 g: FC + PDB 25 g, GNSS 14 g)
Cost allocation	£63 (SpeedyBee F405 WING APP £45, GNSS £14, microSD £4)
Power	≈ 0.8 W including GNSS

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	4	-	-	-	20	23
BENCH	10	-	5	1	-	16

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
POOL	3	-	-	-	1	4
LAKE	1	-	-	-	-	1

Ref	Open item
V-01	Bench check of the SpeedyBee with Rover 4.7.1 (every feature was exercised in SITL)
V-02 ... V-06, V-11 ... V-17	Answered in SITL (ADD section 9); results folded into the requirements below
Tuning	MOT_THST_EXPO, CRUISE_THROTTLE, steering gains from thrust-stand and pool data (HLM-D42)
HLM-OI-1 / CR-02	Closed: owner chose a flight controller with microSD (ADD DD-15)